# ELEMENT COMBINATION RULE

## CIRCUIT ANALYSIS METHOD

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# TOPIC OUTLINE

Ohm's Law

Series Network

Parallel Network

Series-Parallel Network

# OHM'S LAW

# OHM'S LAW

**Ohm's Law** states that the ratio of voltage ($\boldsymbol{V}$) to current ($\boldsymbol{I}$) is **constant** ($\boldsymbol{R}$).

Mathematical Representation

$$\boldsymbol{R = \frac{V}{I}}$$

Basic Electrical Quantities

1. **Voltage (V)**

The measure of electrical potential energy per unit charge. It is the "**push**" or "force" that drives electric current through a circuit.

Formula

$$\boldsymbol{V = IR}$$

unit

Volt (V)

# OHM'S LAW

Ohm's Law states that the ratio of voltage ($V$) to current ($I$) is **constant** ($R$).

Mathematical Representation

$$R = \frac{V}{I}$$

Basic Electrical Quantities

## 2. Current (I):

The **flow of electric charge**, typically carried by electrons in a conductor. It represents the rate at which charge flows through a point in a circuit.

Formula

$$I = \frac{V}{R}$$

unit

Ampere (A)

# OHM'S LAW

Basic Electrical Quantities

3. **Resistance (R)**

The **opposition** to the flow of electric current in a material or component. It determines how much current will flow for a given voltage.

Formula

$$\boldsymbol{R = \frac{V}{I}}$$

unit

Ohm ( Ω )

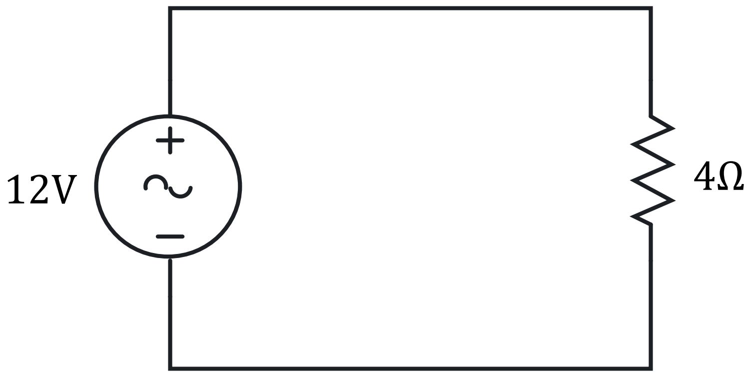
**Ohm's Law** states that the ratio of voltage ($\boldsymbol{V}$) to current ($\boldsymbol{I}$) is **constant** ($\boldsymbol{R}$).

Mathematical Representation

$$\boldsymbol{R = \frac{V}{I}}$$

# EXERCISE

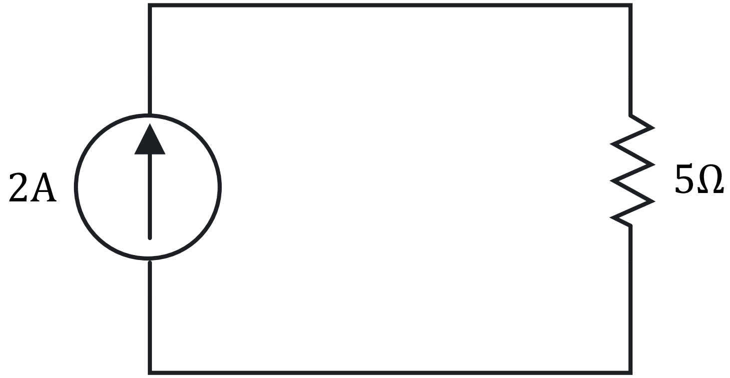
Determine the current flowing through a circuit with 4 ohms resistance when a voltage of 12 volts is applied.




Solution

## EXERCISE

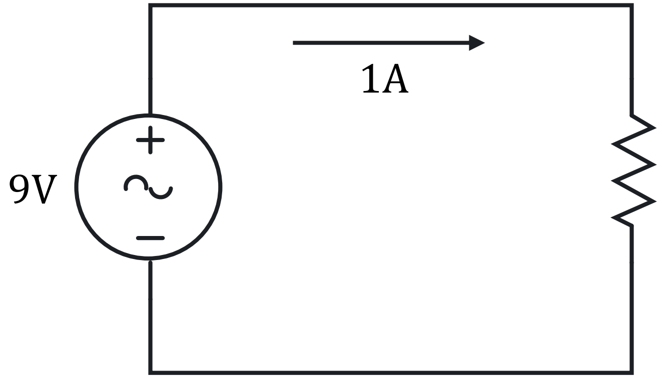
Determine the voltage drop across a circuit with a resistance of 5 ohms when a current of 2 amps flows through it.




Solution

# EXERCISE

Determine the resistance of a circuit if a 9-volt applied voltage results in a current flow of 1-ampere.



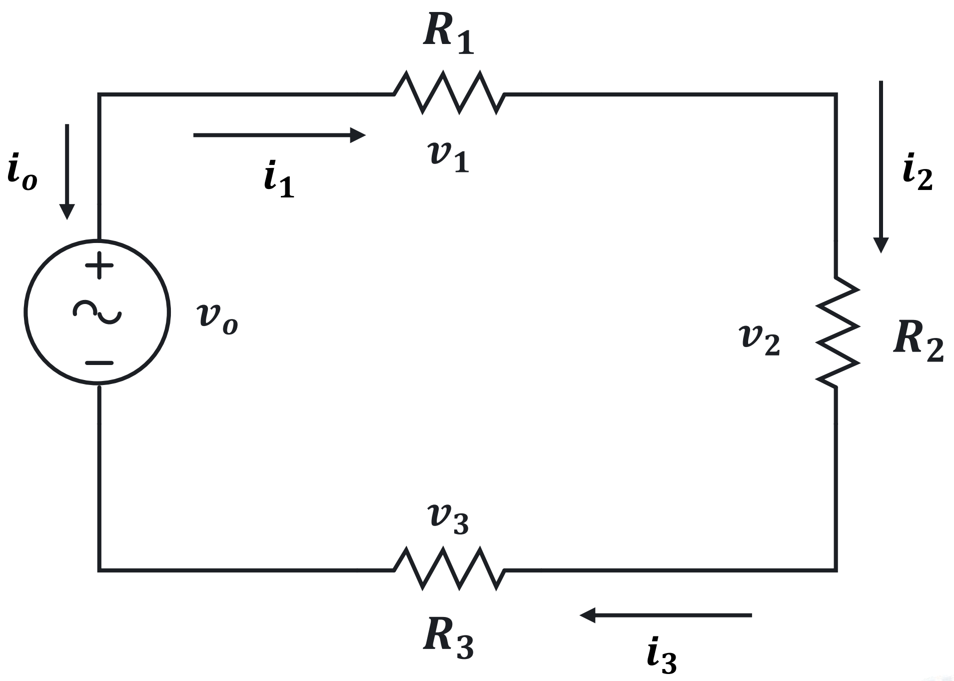

Solution

# SERIES NETWORK

# SERIES NETWORK

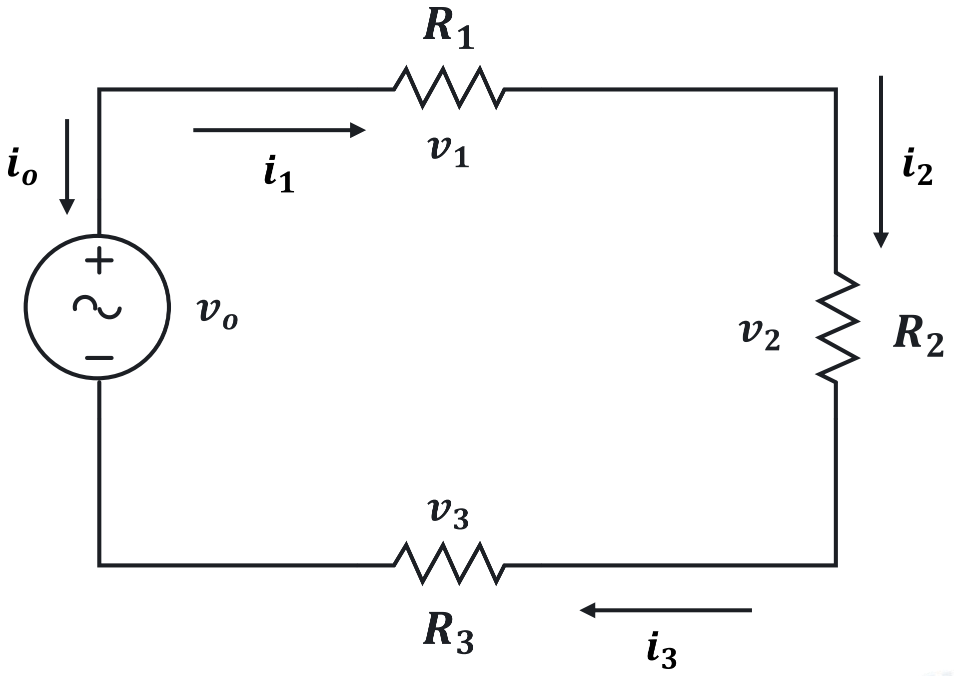
A **series network** refers to a configuration where components are connected end-to-end, forming a **single path** for current to flow.

Series Network



# CURRENT

Series Network



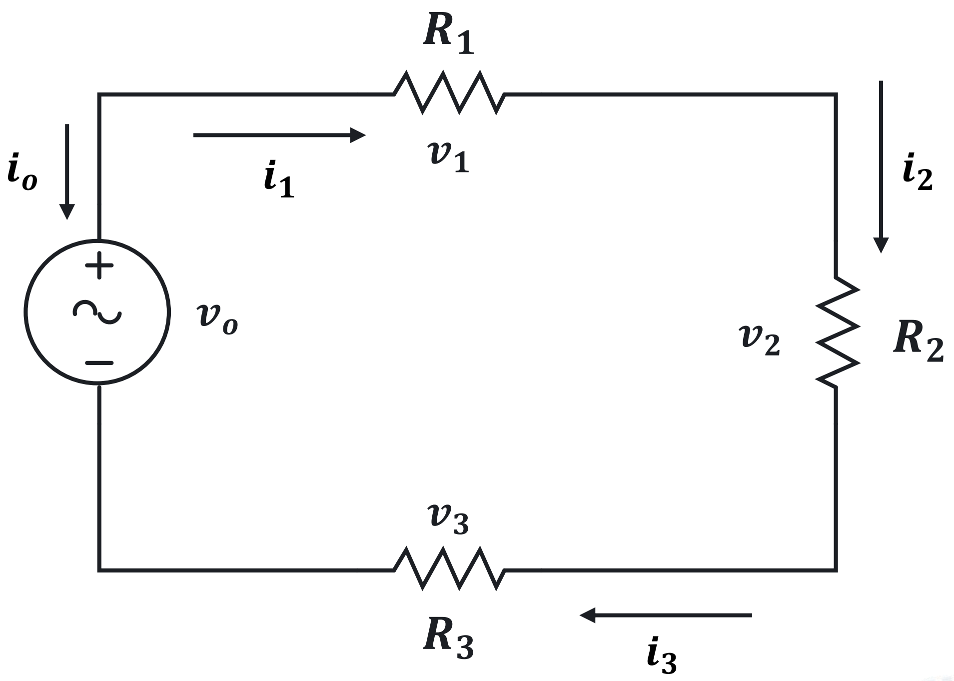

In a series network, the **same current** flows through all components.

Mathematical Representation

$$i_o = i_1 = i_2 = i_3 = \cdots i_n$$

# RESISTANCE

Series Network




In a series network, the **total resistance** is the **sum** of the individual resistances.

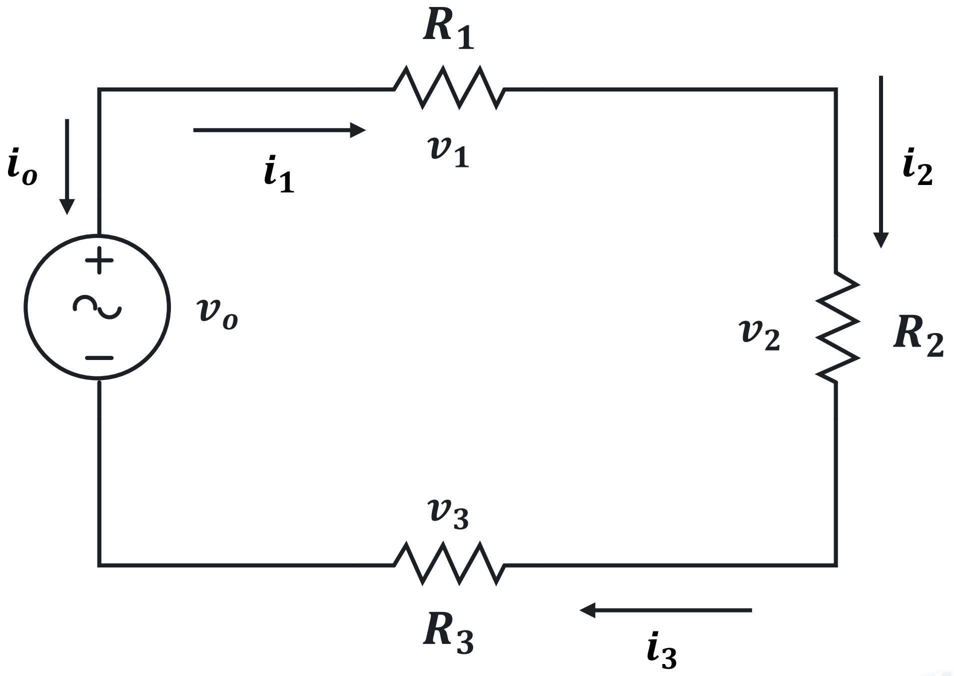
Mathematical Representation

$$R_o = R_1 + R_2 + R_3 + \cdots R_n$$

# VOLTAGE

Series Network

In a series network, the **total voltage** is the **sum** of the voltages across each individual component.

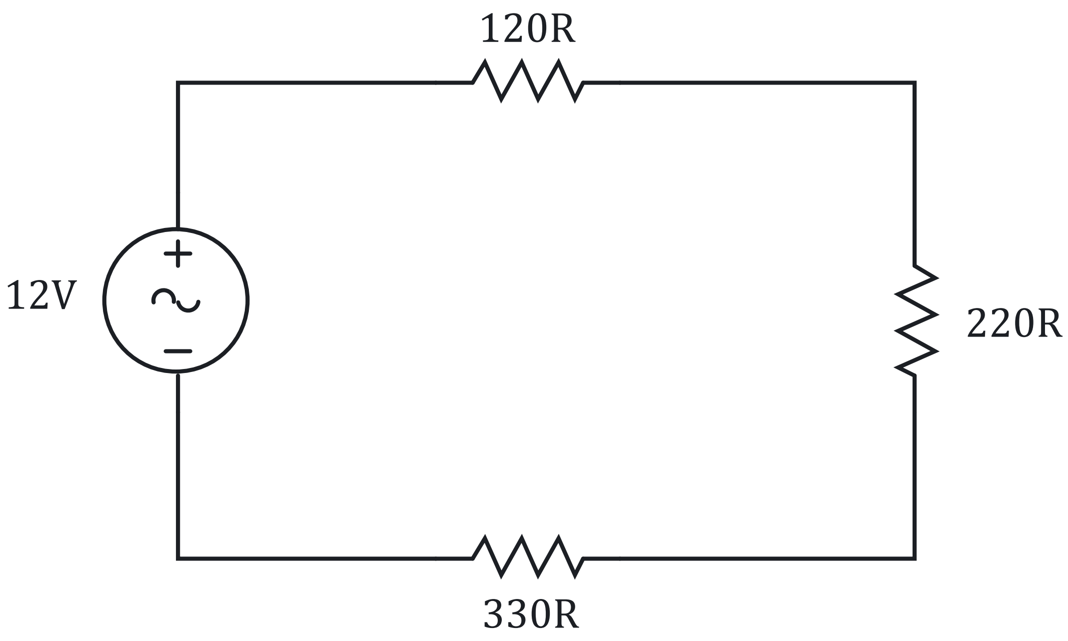



Mathematical Representation

$$v_o = v_1 + v_2 + v_3 + \cdots v_n$$

# EXERCISE

Determine the voltage drop across each resistor of the given circuit.

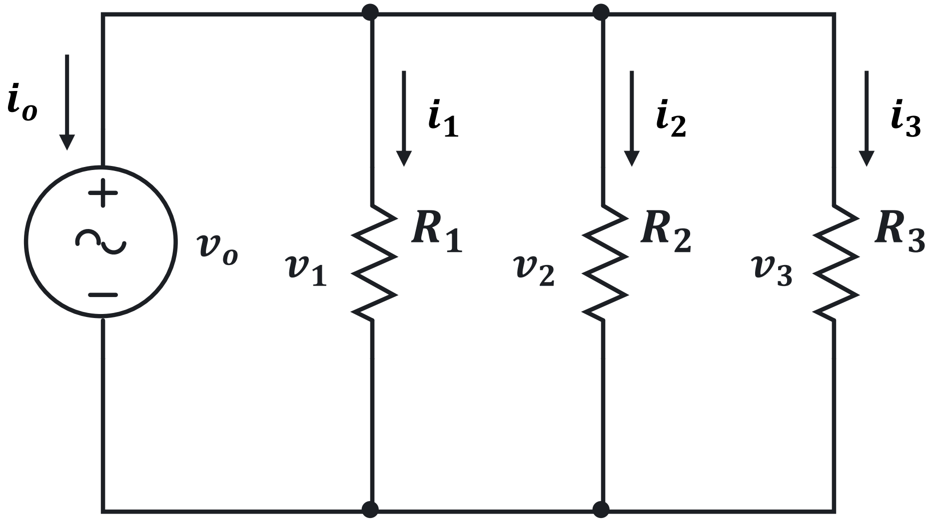



Solution

# PARALLEL NETWORK

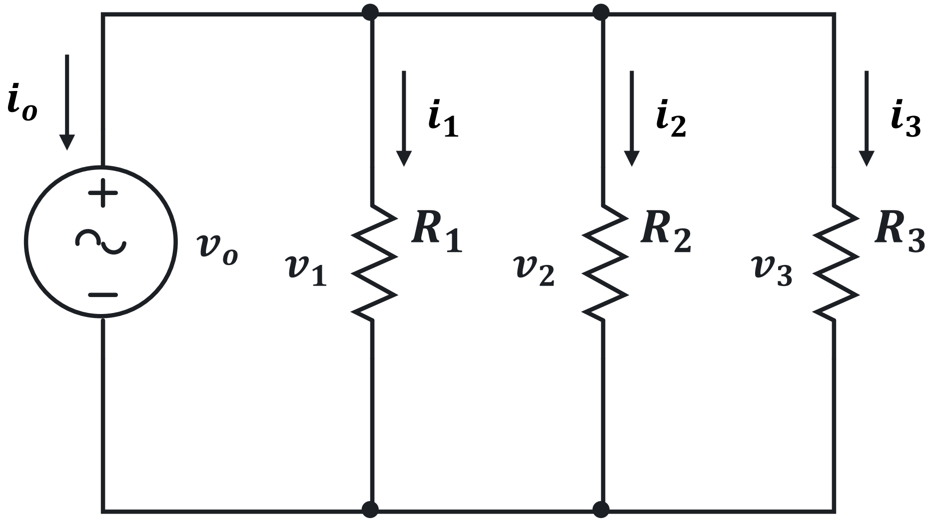
Parallel Network

A **parallel network** is a configuration where components are connected across the **same two points**, providing multiple paths for current to flow.



# VOLTAGE

Parallel Network



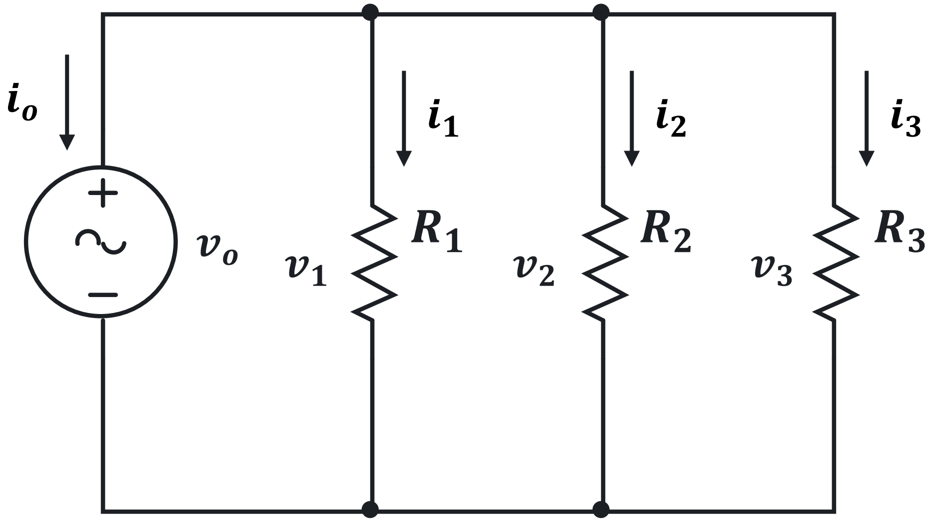

In a parallel network, the voltage across all components is the **same**.

Mathematical Representation

$$v_o = v_1 = v_2 = v_3 = \cdots v_n$$

# CONDUCTANCE

Parallel Network



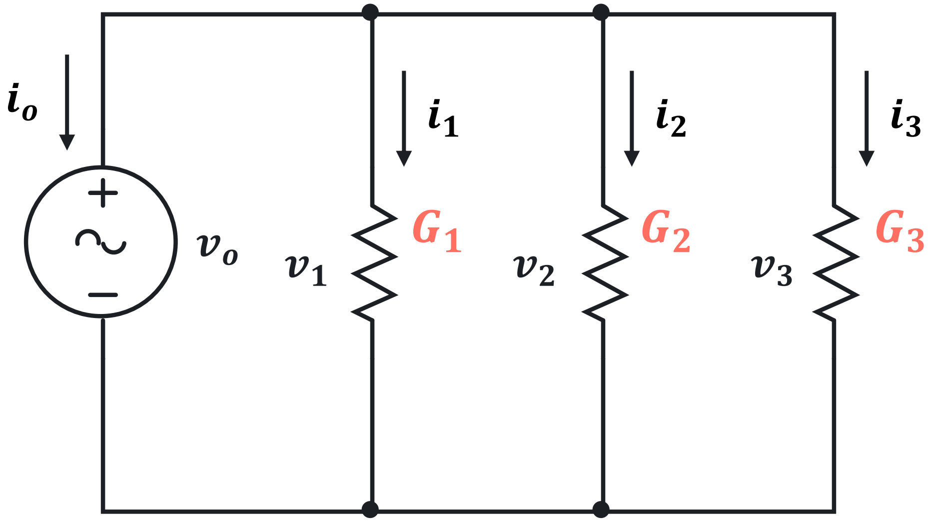

**Conductance** refers to the ability of the network to allow the flow of electric current. It is the **reciprocal of resistance** and is measured in siemens ($\boldsymbol{S}$).

Mathematical Representation

$$G = \frac{1}{R}$$

# CONDUCTANCE

Parallel Network



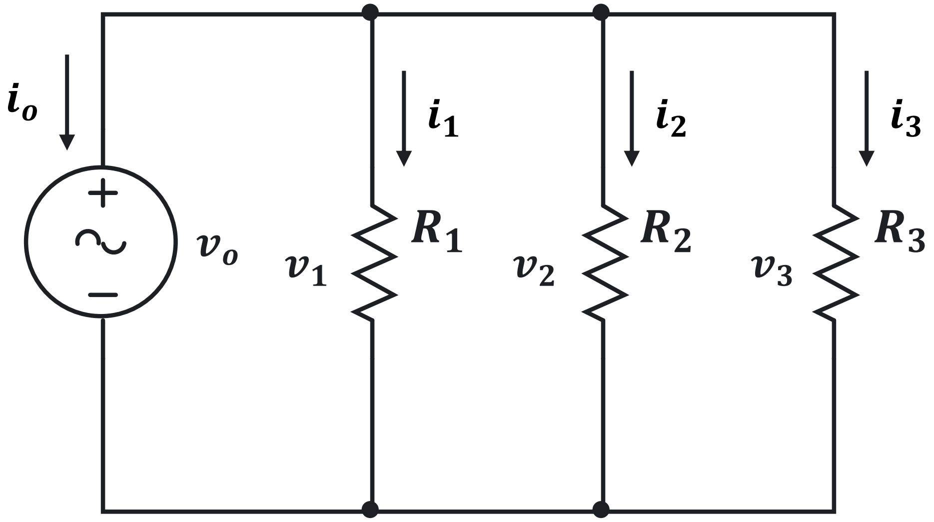

In a parallel network, the **total conductance** is the **sum** of the individual conductance of each resistor.

Mathematical Representation

$$G_o = G_1 + G_2 + G_3 + \cdots G_n$$

# CURRENT

Parallel Network



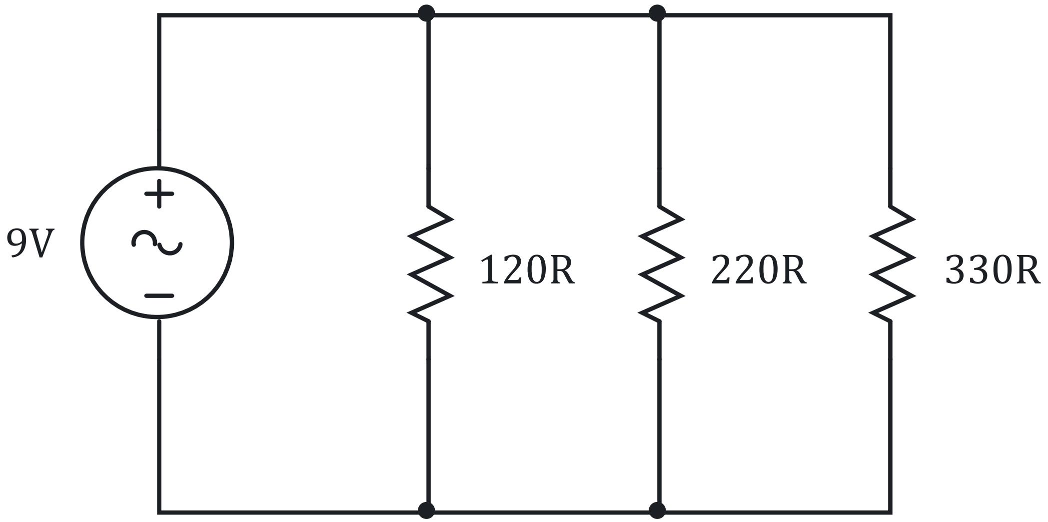

In a parallel network, the **total current** is the **sum** of the current flowing through each individual component.

Mathematical Representation

$$i_o = i_1 + i_2 + i_3 + \cdots i_n$$

## EXERCISE

Determine the current flowing through each resistor of the given circuit.




Solution

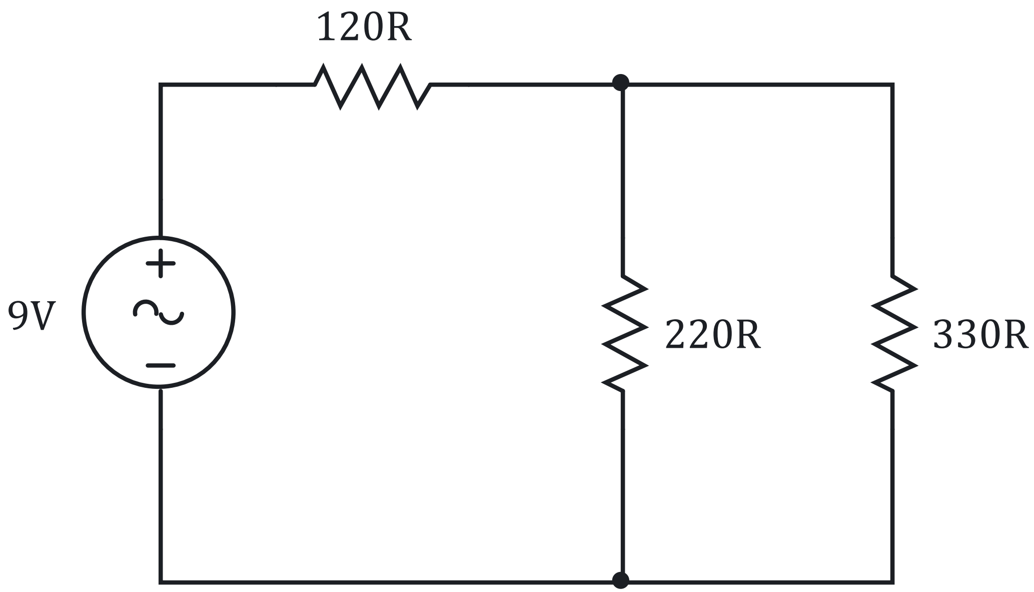
# SERIES PARALLEL NETWORK

# SERIEL-PARALLEL NETWORK

A **series-parallel** network is a type of electrical network that combines elements of both **series** and **parallel** circuits. These networks are commonly used in electrical and electronic systems to achieve desired voltage, current, and resistance characteristics.

## EXERCISE

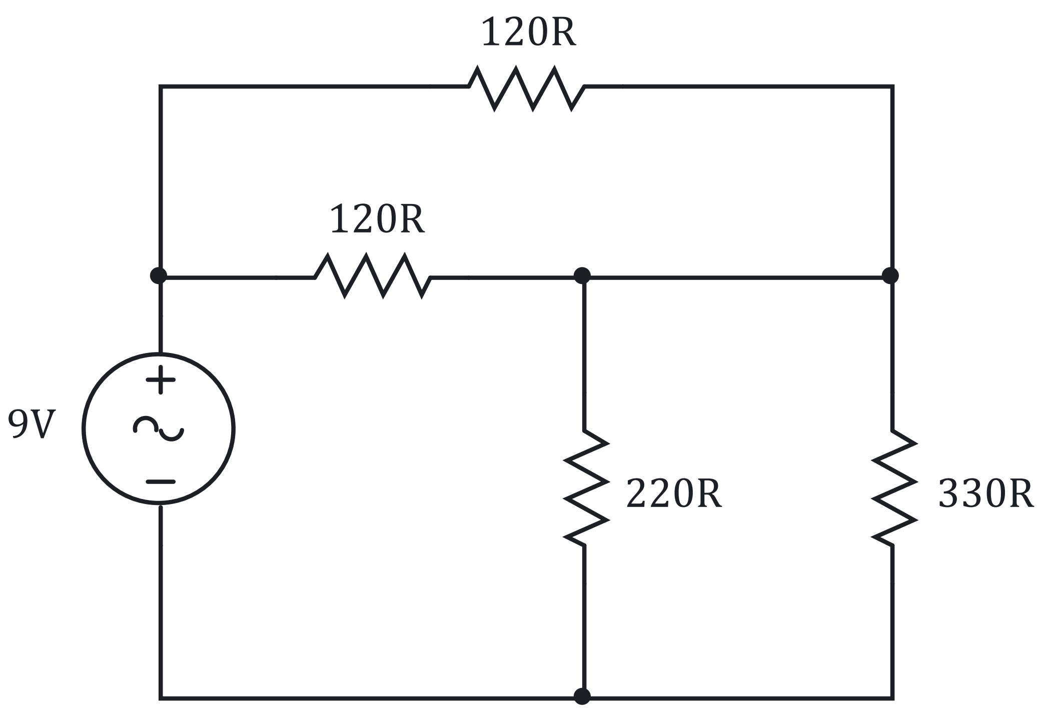
Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.




Solution

# EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.

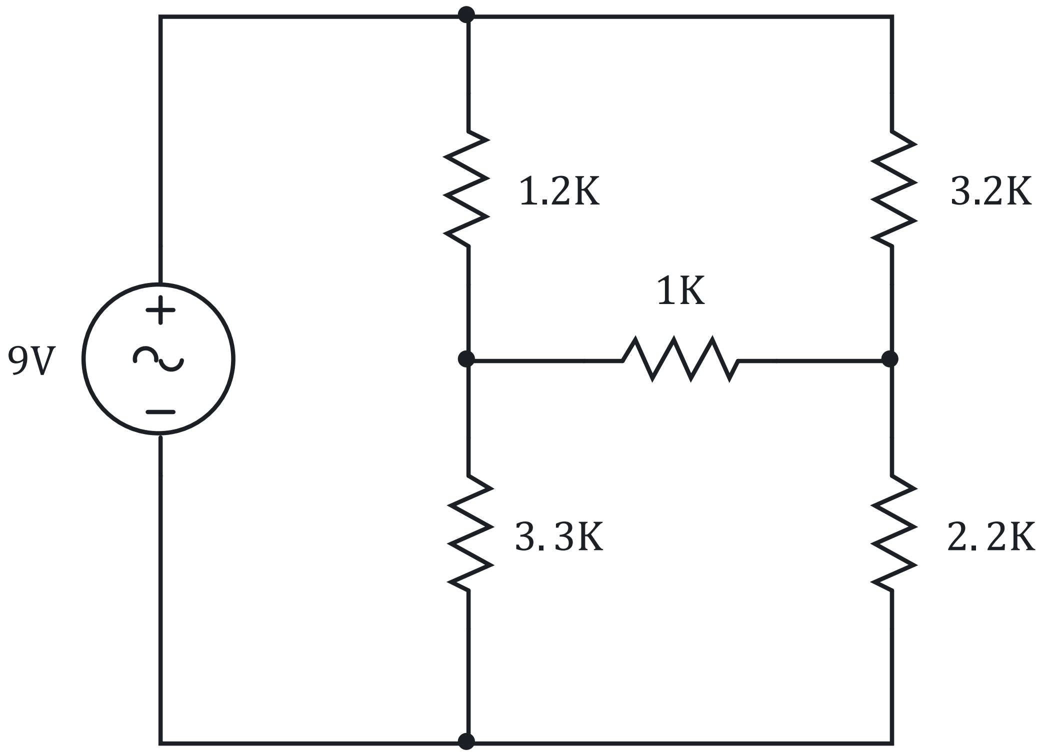



Solution

## EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.

Solution



# LABORATORY